# COHEN SECTIONS FOR CHARACTERISTIC P LOCAL RINGS WITH NILPOTENT MAXIMAL IDEAL, WITH COMPATIBLE VARIANTS

THE STACKS PROJECT AUTHORS

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*Allowed standard prerequisites (editorial).*

- A separable field extension is formally smooth, so a ring map into a quotient with nilpotent kernel lifts through the quotient. invoked at proof lines 34757-34760. Provides the section sigma and its compatibility.

*Source reference.* Original proposition 0322: sources/stacks-chow/tags.txt, source lines 4005–4005. Cited at proof lines 34757-34758 for the formal smoothness of the separable extensions; the conclusion is used inside the counted span.

**Lemma 0.2.** *Let $A$ be a local ring annihilated by a prime $p$ whose maximal ideal is nilpotent. There exists a ring map $\sigma : \kappa_A \to A$ which is a section to the residue map $A \to \kappa_A$. If $A \to A'$ is a local homomorphism of local rings, then we can choose a similar ring map $\sigma' : \kappa_{A'} \to A'$ compatible with $\sigma$ provided that the extension $\kappa_{A'}/\kappa_A$ is separable.*

*Proof.* Separable extensions are formally smooth by Algebra, Proposition 0322. Thus the existence of $\sigma$ follows from the fact that $\mathbf{F}_p \to \kappa_A$ is separable. Similarly for the existence of $\sigma'$ compatible with $\sigma$. □

*Date*: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.